

CampusEats — Assignment 3

Integrate an External SOAP Partner

Team ID: CampusEats-MultiAgent-01

Team Members

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1. Context

CampusEats integrates an external Payment Gateway using the single operation charge. The rest of CampusEats can remain REST-based, while this financial integration is modeled with SOAP because SOAP provides a strong WSDL contract, structured faults and a clear message-oriented security boundary. This is design-time only; no live gateway or server is implemented.

2. WSDL Contract

The supplied partner.wsdl contains types, messages, portType, SOAP-over-HTTP binding, service and port. It defines the charge operation.

Element	Definition
Operation	charge
Input	merchantId, orderId, amount, currency, paymentToken
Output	status, transactionId, orderId, message
SOAPAction	http://campuseats.example.com/payment/charge
Endpoint	https://payments.partner.example.com/soap/PaymentGateway
Binding	document/literal SOAP over HTTP

3. SOAP Messages

Request — credentials are in Header and charge is in Body.

Request.

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:pay="http://campuseats.example.com/payment">
  <soap:Header><pay:Credentials>
    <pay:merchantId>CE-CAMPUS-001</pay:merchantId><pay:apiKey>REDACTED-DEMO-KEY</pay:apiKey>
    <pay:timestamp>2026-08-26T22:30:00+05:30</pay:timestamp>
  </pay:Credentials></soap:Header>
  <soap:Body><pay:ChargeRequest>
    <pay:merchantId>CE-CAMPUS-001</pay:merchantId><pay:orderId>CE-ORD-10045</pay:orderId>
    <pay:amount>249.00</pay:amount><pay:currency>INR</pay:currency><pay:paymentToken>tok_demo_7F83A2</pay:paymentToken>
  </pay:ChargeRequest></soap:Body>
</soap:Envelope>
```

Success response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" xmlns:pay="http://campuseats.example.com/payment">
  <soap:Header/><soap:Body><pay:ChargeResponse>
    <pay:status>APPROVED</pay:status><pay:transactionId>TXN-20260826-88421</pay:transactionId>
    <pay:orderId>CE-ORD-10045</pay:orderId><pay:message>Payment approved</pay:message>
  </pay:ChargeResponse></soap:Body>
</soap:Envelope>
```

Fault response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Header/><soap:Body><soap:Fault>
    <faultcode>soap:Client</faultcode><faultstring>Payment declined</faultstring>
    <detail><PaymentFault><code>card_declined</code><orderId>CE-ORD-10045</orderId>
      <message>Issuer declined the payment.</message></PaymentFault></detail>
    </soap:Fault></soap:Body>
</soap:Envelope>
```

4. HTTP Binding

```
POST /soap/PaymentGateway HTTP/1.1
Host: payments.partner.example.com
Content-Type: text/xml; charset=utf-8
SOAPAction: "http://campuseats.example.com/payment/charge"
https://payments.partner.example.com/soap/PaymentGateway
<soap:Envelope> ... </soap:Envelope>
```

5. Discovery

CampusEats discovers the partner through a modern catalogue/registry entry. No UDDI server is required. The catalogue points the consumer to the partner WSDL.

Field	Registry entry
business	CampusPay External Payment Services
service	CampusEats Payment Gateway
endpoint	https://payments.partner.example.com/soap/PaymentGateway
WSDL / tModel-style pointer	https://payments.partner.example.com/wsdl/PaymentGateway.wsdl
operation	charge

6. Fault Mapping

The partner fault `card_declined` is mapped internally to the CampusEats contract error `placeOrder` → payment declined. The partner vocabulary is not exposed to students. The student receives a stable message such as: Payment could not be completed for this order.